



JOHNS HOPKINS
UNIVERSITY

EN.601.422 / EN.601.622

Software Testing & Debugging

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Logic Coverage

- ▶ With *equivalence partitioning*, we make certain we adequately cover the domain of input/output values
- ▶ With Whitebox coverages (statement- & branch-coverages) we make sure we reach all different parts and branches of the code
- ▶ With *logic coverage*, we focus on the combinations of truth assignments of the logical expressions in loops and conditionals

Logic Coverage Terminology

- ▶ **Predicate:** an expression that evaluates to either true or false
 - ❖ Predicates contain Boolean variables, relational operators (e.g., $<$, $>$, $!=$, $==$ etc.), Boolean function calls, logical operators (e.g., logical AND, logical OR etc.) or Boolean values (i.e., true and false)
- ▶ **Clause:** A predicate with no logical operators
- ▶ **Logical Operators:**
 - ❖ $\neg$ – the *negation* operator
 - ❖ $\wedge$ – the *and* operator
 - ❖ $\vee$ – the *or* operator
 - ❖ $\rightarrow$ – the *implication* operator
 - ❖ $\oplus$ – the *exclusive or* operator
 - ❖ $\leftrightarrow$ – the *equivalence* operator

Logic Expression

- ▶ Logic expressions may be derived from variety of artifacts:
 - ❖ **Source code:** e.g., `if ((x > 10 && y < -22) || !(z == null))`
 - ❖ **Specifications:** *“to be able to push a new item onto stack, the stack should not be full and stack object not null”*
 - ❖ **FSMs:** “state = card_inserted and action = PIN_ok → state = show_accounts”
 - ❖ **SQL queries:** “SELECT AVG(Salary), Emp_Age FROM Employee WHERE Salary > 100,000 and Emp_Age < 35”
 - ❖ etc.

Logic Expression

► Example: $(a < b) \vee f(z) \wedge D \wedge \text{TRUE}$

❖ $(a < b)$ is relational Boolean expression

❖ $f(z)$ is a Boolean function call (function f returns true or false)

❖ D is a Boolean variable

❖ TRUE is a Boolean value

► The predicate is: “ $(a < b) \vee f(z) \wedge D \wedge \text{TRUE}$ ”

► The four clauses are: “ $(a < b)$ ”, “ $f(z)$ ”, “ D ”, and “ TRUE ”

Logic Expression

- ▶ Human Language can be vague sometimes:
 - ❖ Example: *“I am interested in EN.601.622 and EN.601.682”*
 - Course = *EN.601.622* OR Course = *EN.601.682*
- ▶ Especially important when applying logic coverage on informal specifications, user manual, API docs, etc.

Logic Coverage Criteria

- Assume $\mathbf{P}$ is the set of all predicates and $\mathbf{C}$ is the set of all clauses in P

Predicate Coverage (PC) : For each $p \in P$, TR contains two requirements: p evaluates to true, and p evaluates to false.

Clause Coverage (CC) : For each $c \in C$, TR contains two requirements: c evaluates to true, and c evaluates to false.

Example

► $(\underbrace{x > y}_{C1} \vee \underbrace{f(z)}_{C2}) \wedge \underbrace{w}_{C3}$

► **Predicate Coverage:**

- ❖ $x = 10, y = 9, f(z) = \text{true}, w = \text{true}$
- ❖ $x = 5, y = 5, f(z) = \text{false}, w = \text{true}$

the predicate evaluates to true
the predicate evaluates to false

► **Clause Coverage:**

- ❖ $x = 10, y = 9, f(z) = \text{true}, w = \text{true}$
- ❖ $x = 5, y = 5, f(z) = \text{false}, w = \text{false}$

C1 is true, C2 is true, C3 is true
C1 is false, C2 is false, C3 is false

Logic Coverage Criteria

- ▶ *PC* does not fully exercise all the clauses, especially in the presence of short circuit evaluation
- ▶ **Short circuit:** Not all clauses may be evaluated
 - ❖ Example 1: $x > y \vee f(z)$ $f(z)$ ignored if $\mathbf{x} > \mathbf{y}$ is true
 - ❖ Example 2: $w \wedge f(z)$ $f(z)$ ignored if $\mathbf{w}$ is false
- ▶ **CC** does not subsume **PC**, i.e., we may satisfy CC without causing the predicate to be both true and false
- ▶ The most comprehensive solution is to try all possible combinations of the clauses

Logic Coverage Criteria

- ▶ Assume $\mathbf{P}$ is the set of all predicates and $\mathbf{C}$ is the set of all clauses in $\mathbf{P}$

Combinatorial Coverage (CoC) : For each $p \in P$, TR has test requirements for the clauses in C_p to evaluate to each possible combination of truth values.

Example

$$(x > y \vee f(z)) \wedge w$$

c1

c2

c3

C1	C2	C3	P
T	T	T	T
T	T	F	F
T	F	T	T
T	F	F	F
F	T	T	T
F	T	F	F
F	F	T	F
F	F	F	F

C1	C2	C3	P
x = 1, y = 0	f(z) = true	w = true	T
x = 1, y = 0	f(z) = true	w = false	F
x = 1, y = 0	f(z) = false	w = true	T
x = 1, y = 0	f(z) = false	w = false	F
x = 1, y = 1	f(z) = true	w = true	T
x = 1, y = 1	f(z) = true	w = false	F
x = 1, y = 1	f(z) = false	w = true	F
x = 1, y = 1	f(z) = false	w = false	F

Combinatorial Coverage Criterion

- ▶ For a predicate with n clauses, there are 2^n possible truth assignments
- ▶ Not scalable, maybe even impractical if number of clauses is very large

****Is there a way to capture the *effect* of each and every clause while not exhausting all combinations?****

Active Clause

Active Clause

- ▶ Can we control for all the clauses in a predicate except one such that that one clause (i.e., active/major clause) would decide the outcome of the predicate?
 - ❖ **Example:** $(a \vee b) \wedge c$ **a** is “active” if **b** is false and **c** true
 c is active if $(a \vee b)$ is true

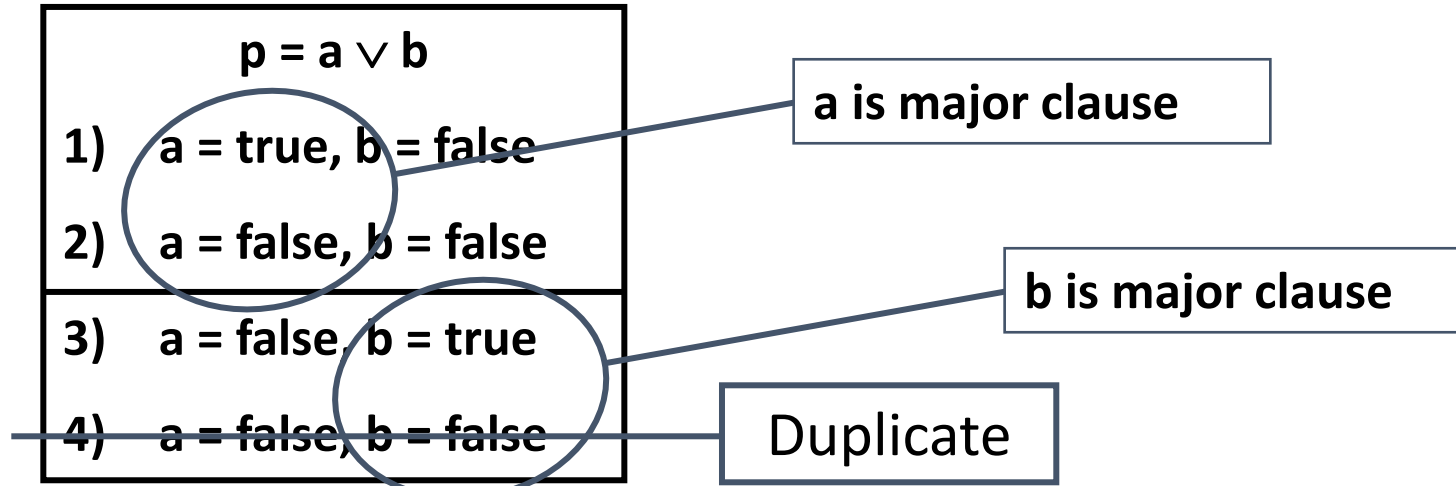
Determination	A clause C_i in predicate p , called the <i>major clause</i> , <i>determines</i> p if and only if the values of the remaining <i>minor clauses</i> $C_j, (j \neq i)$, are such that changing C_i changes the value of p
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Determination

- ▶ Our goal is to find tests for each clause when the clause determines the value of the predicate
- ▶ An important thing to note about “determination”
 - ❖ the definition does not require $C_i = p$ (C_i can be negation of p)
 - *Example:* $a \leftrightarrow b$ a is active if b is false
 $a = \text{false}, b = \text{false} \rightarrow P = \text{true}$

Active Clause Coverage

Active Clause Coverage (ACC) : For each $p \in P$ and each major clause C_i in C_p , choose minor clauses $C_j, j \neq i$, so that C_i determines p . TR has two requirements for each C_i : C_i evaluates to true and C_i evaluates to false.



General Active Clause Coverage

General Active Clause Coverage (GACC) : For each p in P and each major clause c_i in C_p , choose minor clauses $c_j, j \neq i$, so that c_i determines p . TR has two requirements for each c_i : c_i evaluates to true and c_i evaluates to false. The values chosen for the minor clauses c_j do not need to be the same when c_i is true as when c_i is false.

Example

$$p = a \leftrightarrow b$$

- ▶ **a** determines **p** no matter what **b**'s value is
 - ❖ Assume **b** is false. Now, if **a** is true $\rightarrow p = \text{false}$ and if **a** is false $\rightarrow p = \text{true}$
 - ❖ Assume **b** is true. Now, if **a** is true $\rightarrow p = \text{true}$ and if **a** is false $\rightarrow p = \text{false}$
- ▶ Likewise, **b** determines **p** no matter what **a**'s value is
 - ❖ Assume **a** is false. Now, if **b** is true $\rightarrow p = \text{false}$ and if **b** is false $\rightarrow p = \text{true}$
 - ❖ Assume **a** is true. Now, if **b** is true $\rightarrow p = \text{true}$ and if **b** is false $\rightarrow p = \text{false}$
- ▶ TR contains both **a** and **b** evaluate to both true and false
 - ❖ Can be achieved by {TT, FF}: TT $\rightarrow p = \text{true}$
FF $\rightarrow p = \text{true}$

General Active Clause Coverage

- ▶ It is possible to satisfy GACC without satisfying predicate coverage
- ▶ GACC does not subsume predicate coverage (PC)
- ▶ We really want to cause predicates to be both true and false!

Correlated Active Clause Coverage

Correlated Active Clause Coverage (CACC) : For each p in P and each major clause c_i in C_p , choose minor clauses $c_j, j \neq i$, so that c_i determines p . TR has two requirements for each c_i : c_i evaluates to true and c_i evaluates to false. The values chosen for the minor clauses c_j must cause P to be true for one value of the major clause c_i and false for the other, that is, it is required that $p(c_i = \text{true}) \neq p(c_i = \text{false})$.

Example

$$p = a \leftrightarrow b$$

- ▶ **a** determines **p** no matter what **b**'s value is
 - ❖ Assume $b = \text{false}$. Then $a = \text{true} \rightarrow p = \text{false}$ and $a = \text{false} \rightarrow p = \text{true}$
 - ❖ Assume $b = \text{true}$. Then $a = \text{true} \rightarrow p = \text{true}$ and $a = \text{false} \rightarrow p = \text{false}$
- ▶ **b** determines **p** no matter what **a**'s value is
 - ❖ Assume $a = \text{false}$. Then $b = \text{true} \rightarrow p = \text{false}$ and $b = \text{false} \rightarrow p = \text{true}$
 - ❖ Assume $a = \text{true}$. Then $b = \text{true} \rightarrow p = \text{true}$ and $b = \text{false} \rightarrow p = \text{false}$
- ▶ TR should contain both **a** and **b** evaluate to both true and false and **p** must evaluate to true and false for each truth assignment of either **a** or **b**.
 - ❖ CACC CanNOT be achieved by $\{TT, FF\}$:
 $TT \rightarrow p = \text{true}$
 $FF \rightarrow p = \text{true}$

Example

$$p = a \leftrightarrow b$$

► CACC can now be achieved by the combinations in the table:

- ❖ a is the major clause:
 - a = true, p = true (1st row)
 - a = false, p = false (3rd row)
- ❖ b is the major clause:
 - b = true, p = true (1st row)
 - b = false, p = false (2nd row)

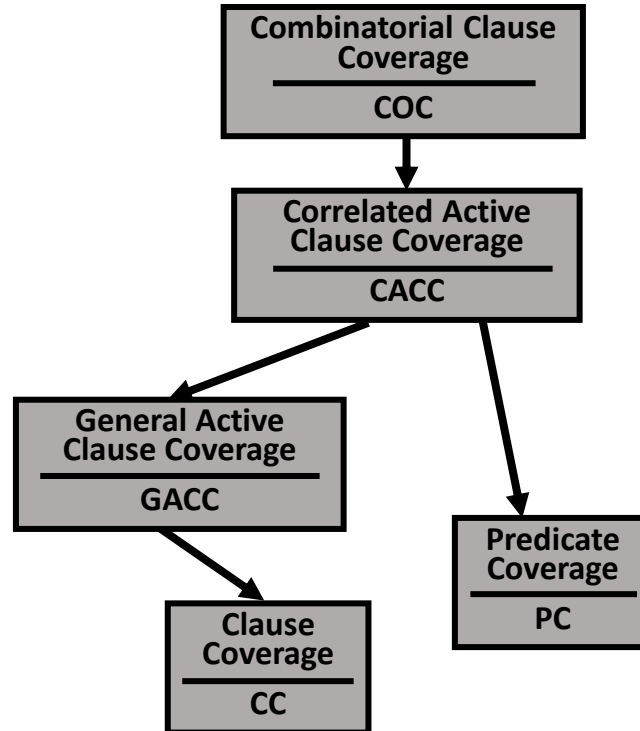
a	b	p
T	T	T
T	F	F
F	T	F

Difference between CACC and GACC

► The core difference:

- ❖ **GACC** only requires that changing the major clause can change the predicate
- ❖ **CACC** requires showing that the major clause can make the predicate **both true and false**.

Logic Criteria Subsumption



Exercise

- ▶ First, formulate the following sentence as a logic predicate:
 - ❖ *“List all Canon (the brand) printers in the store with a price of greater than \$300 or for which the store has more than 100 items. Also, list all non-Canon printers with price less than \$200”*
- ▶ Next, write truth assignments to achieve:
 - ❖ *Clause Coverage*
 - ❖ *CACC*

$\text{Canon}(e) \wedge (\text{price}(e) > 300 \vee \text{count}(e) > 100) \vee (\text{non-Canon}(e) \wedge \text{price}(e) < 200)$

Clause Coverage

C1	C2	C3	C4	C5	P
T	T	T	F	T	T
F	F	F	T	F	F

Constraint: $C1 \oplus C4$

CACC

C1	C2	C3	C4	C5	P
T	T	-	F	-	T
F	T	-	T	F	F
T	T	F	F	F	T
T	F	F	F	F	F
T	F	T	F	F	T
T	F	F	F	F	F
F	-	-	T	T	T
F	-	-	F	T	F
F	-	-	T	T	T
F	-	-	T	F	F